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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

EC2105-1 – NETWORK THEORY AND CONTROL SYSTEMS
 (For ECE, ACT, VDT)

Duration: 3 Hours

Max. Marks: 100

Note:

Usage of Graph sheet and Semilog Sheet is permitted

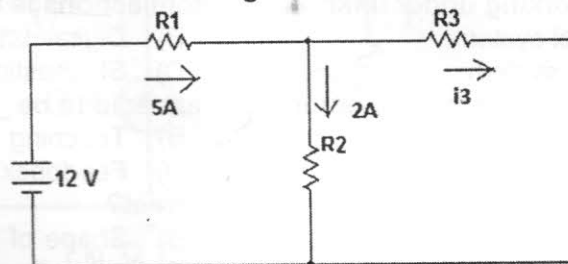
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which law plays a significant role in the mesh analysis of the network?
 A) KVL
 B) KCL
 C) Law of superposition
 D) Law of Reciprocity
- Find the current i_3 in the circuit shown in Fig.



- Find the current i_3 in the circuit shown in Fig.
 A) 5A
 B) 2A
 C) 3A
 D) 1A
- A star connection has each element of resistance $\frac{R}{2}$. The equivalent resistance in delta connection will be
 A) $\frac{R}{6}$
 B) $\frac{3R}{2}$
 C) $2R$
 D) $4R$
- Potential difference in electrical terminology is known as?
 A) Voltage
 B) Resistance
 C) Current
 D) Conductance
- Applying Norton's theorem to a linear two-terminal network yields an equivalent circuit consisting of
 A) A single equivalent current source and an impedance in series
 B) A single equivalent current source and an impedance in parallel
 C) A single equivalent impedance
 D) A single equivalent voltage source and an impedance in parallel
- Thevenin's voltage is equal to the _____ voltage across the _____ terminals.
 A) open circuit, output
 B) open circuit, input
 C) short circuit, input
 D) short circuit, output

- 2-

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

Marks BT* CO* PO*

1. a) For the circuit shown in Fig. Q1 (a), find i_1 using source transformation.

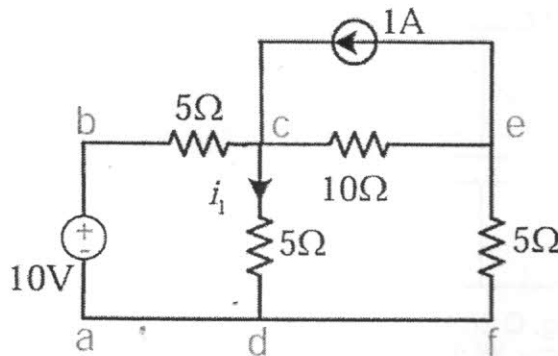


Fig. Q1 (a)

8 L*3 1 1

- b) Using superposition theorem, find the value of I for the circuit shown in Fig. Q 1(b).

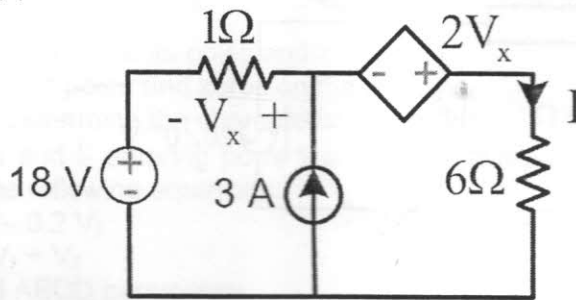


Fig. Q 1(b)

8 L3 2 1

2. a) Using mesh current analysis, find the current through $10\ \Omega$ resistor for the circuit shown in Fig. Q2 (a).

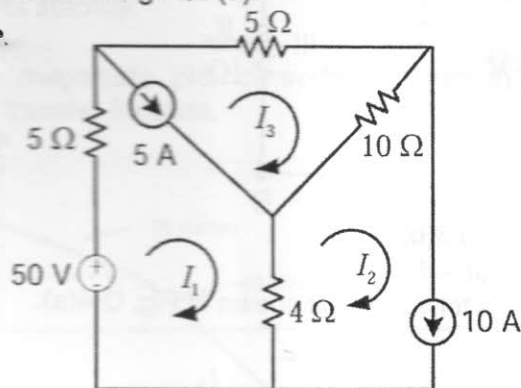


Fig. Q2 (a)

8 L3 1 1

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- b) What should be the value of the pure resistance to be connected across the terminals A – B in the network shown in Fig. Q 2(b) so that the maximum power is transferred to the load? What is the maximum power?

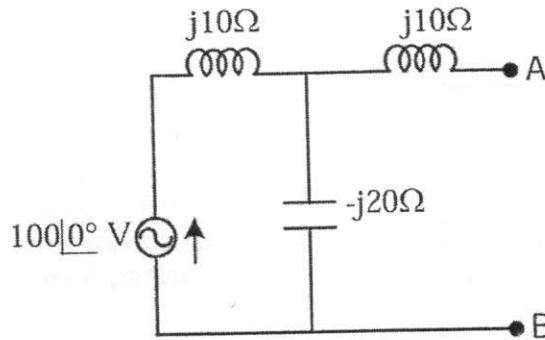


Fig. Q 2(b)

8 L3 2 1

3. a) Consider the circuit shown in Fig. Q 3(a). Find the value of I using node voltage analysis.

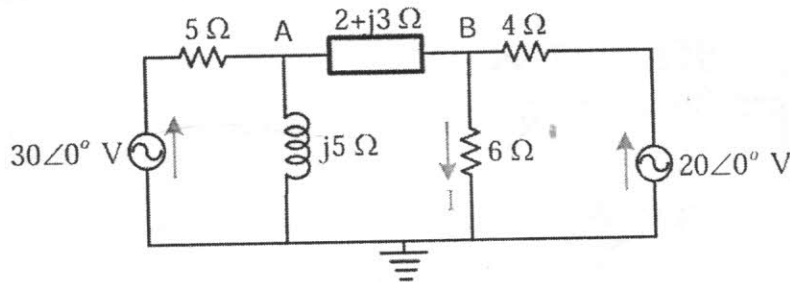


Fig. Q 3(a)

8 L3 1 1

- b) Calculate the current I_L flowing through the load R_L for the circuit shown in Fig. Q 3(b) using Norton's theorem.

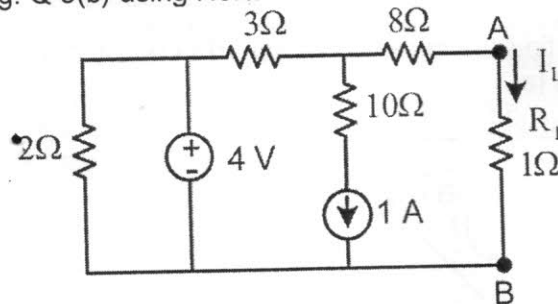


Fig. Q 3(b)

8 L3 2 1

Unit – II

4. a) Find the transmission parameters for the circuit shown in Fig. Q 4(a).

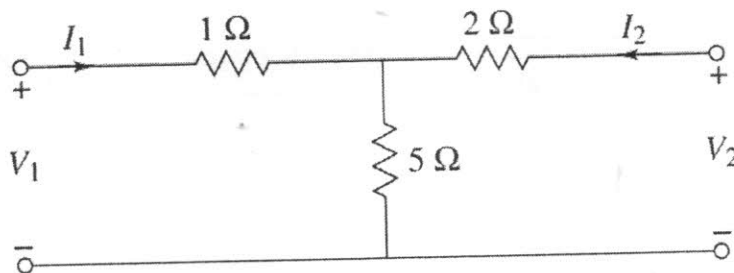


Fig. Q 4(a)

8 L3 3 1

- b) Find the range of values of 'K' so that the system with the following characteristic equation will be stable:
 $s(s^2 + s + 1)(s + 4) + K = 0$
5. a) Consider the Signal Flow Graph shown in Fig. Q 5(a). Find the overall transfer function using Mason's Gain Formula.

8 L3 4 1

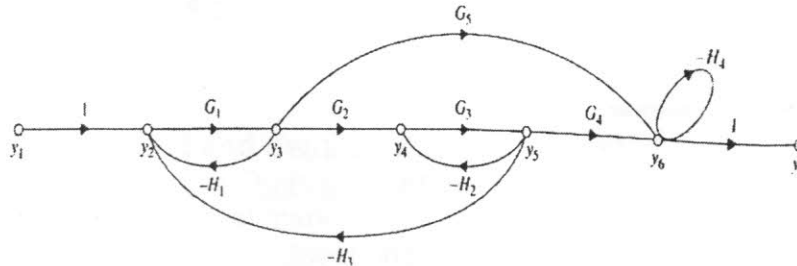


Fig. Q 5(a)

8 L3 4 1

- b) Consider the transfer function of a system given by

$$T(S) = \frac{K(s+6)}{s(s+2)(s+5)(s^2+7s+2)}$$

- (i) Determine its poles and zeros
 (ii) Plot poles and zeros on the S- plane
 (iii) Determine the characteristic equation
6. a) Currents I_1 and I_2 entering ports 1 and 2 of a two-port network are given by the following equations:
 $I_1 = 0.5 V_1 - 0.2 V_2$
 $I_2 = -0.2 V_1 + V_2$
 Find Y and ABCD parameters.
- b) Sketch the complete root locus of a system with

8 L3 4 1

6 L3 3 1

$$G(S)H(S) = \frac{K}{s(s+1)(s+2)(s+3)}$$

Comment on its stability.

10 L3 4 1

Unit – III

7. a) Consider the magnitude plot of system shown in Fig. Q 7(a). Determine its transfer function.

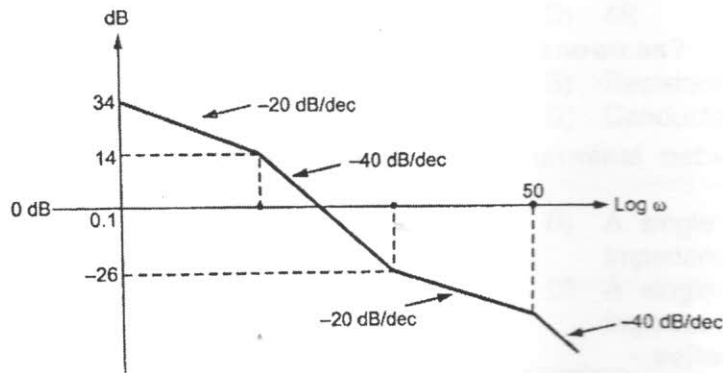


Fig. Q 7(a)

6 L3 5 1

- b) Consider a unity feedback system with $G(S) = \frac{800(s+2)}{s^2(s+10)(s+40)}$

Sketch the Bode plot. Comment on its stability.

10 L3 5 1

8. a) Obtain the transfer function of a unity feedback system whose magnitude plot is as given in Fig. Q 8(a).

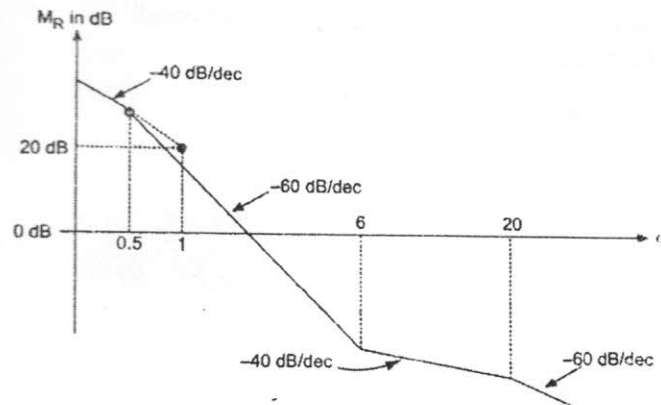


Fig. Q 8(a)

- b) For a unity feedback system has $G(S) = \frac{80}{s(s+2)(s+20)}$

Sketch the Bode plot. Comment on its stability.

6 L3 5 1

10 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
